CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EC) Examination

May 2014

EC312 Digital Signal Processing

Date: 15.05.2014, Thursday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Enlist the Advantages and Applications of Digital Signal Processing. [04]

(b) Determine if the systems described by the following input-output equations are time variant or time invariant.(i) $y(n) = x(2n)$ (ii) $y(n) = (\cos\omega_0 n)x(n)$ (iii) $y(n) = x(n)u(n)$ [03]

Q - 2 (a) Prove Frequency shifting property, Time shifting property and Convolution theorem of Fourier Transform. [07]

(b) Compute the convolution y(n) for the following pair of signals. [07]

$x(n)=\{\underset{\uparrow}{1},1,1,1\}$ and $h(n)=\{6,5,4,\underset{\uparrow}{3},2,1\}$.Sketch x(n),h(n) and y(n).

OR

Q - 2 (a) Draw the block diagram of Digital Signal Processing System. Describe Sampling Process, Aliasing Effect and Reconstruction of band limited signals from their samples. [07]

(b) When the input to linear time invariant system is [07]

$x(n) = 5u(n)$

And output is

$y(n) = [2(0.5)^n + 3(-0.75)^n]u(n)$

(i)Find the H(z) of the system and Plot poles and zeros of H(z).

(ii)Write Difference equation that characterizes the system.

Q - 3 (a) Specify the properties of Region of Conversion (ROC).Prove any two properties of Z-Transform with necessary examples. [07]

(b) Find Inverse Z-Transform of given X[z] using Partial Fraction Expansion Method. [07]

$$X[z] = \frac{z^4}{[z-1][z-0.5]^3}$$

OR

Q - 3 (a) Determine Z-transform and sketch the ROC of following sequences. [07]

(i) $x(n) = [4(5^n) - 9(8^n)]u(n)$

(ii) $x(n) = (\cos\omega_0 n)u(n)$

(iii) $x(n) = \{3, 0, 0, \underset{\uparrow}{1}, 2, 0, 0, 3\}$

(b) Find Inverse Z-Transform of given X[z] using Partial Expansion Method. **[07]**

$$(i) H[z] = \frac{3-4z^{-1}}{\left(1-0.5z^{-1}\right)\left(1-3z^{-1}\right)} \qquad (ii) H[z] = \frac{7-18z^{-1}+6z^{-2}}{\left(1-0.5z^{-1}\right)\left(1-3z^{-1}\right)}$$

SECTION – II

Q - 4 (a) Define Twiddle Factor. Describe Cyclic property of twiddle factor. **[04]**

(b) Explain Frequency Warping of Bilinear Z-Transformation method with necessary figure. **[03]**

Q - 5 (a) List the various methods of designing Infinite Impulse Response Filter. Explain Impulse invariance method. Discuss Pole Mapping Procedure. **[07]**

(b) The Transfer function of discrete time causal system is given below. **[07]**

$$H[z] = \frac{1-\frac{1}{5}z^{-1}}{\left(1-\frac{1}{2}z^{-1}+\frac{1}{3}z^{-2}\right)\left(1+\frac{1}{4}z^{-1}\right)}$$

(i) Draw Direct Form -I Realization

(ii) Draw Direct Form -II Realization

(iii) Draw Cascade Form Realization.

OR

Q - 5 (a) Describe Different Types of Windows functions with mathematical equation and spectrum for designing Finite Impulse Response Filter. Give significance of Kaiser Window. **[07]**

(b) Obtain Parallel Form Realizations for following H[z]. **[07]**

$$\text{(i) } H[z] = \frac{1+0.33z^{-1}}{\left(1-0.5z^{-1}\right)\left(1-0.25z^{-1}\right)} \quad \text{(ii) } H[z] = \frac{1-z^{-1}}{\left(1+0.3z^{-1}\right)\left(1-0.5z^{-1}\right)}$$

Q - 6 (a) If X[n] is given as {1,2,3,5,5,3,2,1}.Find corresponding Discrete Fourier Transform (DFT) of X[k] using Decimation In Time Fast Fourier Transform (DIT FFT) Algorithm. **[07]**

(b) Compute 4 point DFT of following sequences. Use matrix for DFT computation **[07]**

(i) x[n]={1,0,1,0}

(ii) x[n]={0,1,2,3}

OR

Q - 6 (a) Explain Goertzel Algorithm. Determine the length-4 sequence from its DFT $X(k) = \{2, 1-j, 0, 1+j\}$ **[07]**

(b) Define Circular Convolution. Perform Circular Convolution of given two sequences. x[n]={2,0,0,1} and h[n]={4,3,2,1}. **[07]**

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